

Research Statement

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Topic: Vision-based Learning for Swarm Intelligence

Core argument

Vision-based learning can serve as a unifying foundation for swarm intelligence, enabling scalable robot teams to achieve high autonomy and safety in dynamic unknown environments through collaborative visual search, robust decision-making, and vision-language-action (VLA) planning.

➤ Research overview and evidence of impact

Autonomous swarms are moving from controlled demos to real missions (disaster response, low-altitude logistics/traffic), where environments are dynamic and partially observable and where sensing, bandwidth, and onboard compute are limited. The central challenge is safe collaborative decision-making and control for robotic swarms in unknown environments, enabling a swarm to see consistently, *coordinate* efficiently, and *remain resilient* under adversaries and cyber-physical failures. I tackle this challenge through three coherent contributions: **C1 develops vision-based collaboration for unknown-environment search and navigation; C2 studies perception-limited adversarial learning and resilience (pursuit detection/tracking, pursuit-evasion, and formation robustness); and C3 introduces VLA planning to provide structured, controllable guidance for embodied decision-making.** The impact is evidenced by both scholarly and translational signals. I have published over 30 papers (including 16 first-author, with multiple papers in top-tier Journals/conference such as IEEE TNNLS, TRO, CVPR) and obtained 2 international invention patents. My methods have been validated in real-world settings and supported by **>\$3.5M** externally funded projects with Singapore government agencies and industry partners, leading to deployment-oriented deliverables and patents. My research has been cited and positively discussed by leading researchers in robotics and intelligent systems (including multiple academicians and numerous IEEE/AIAA Fellows). They described my collaborative search methods as “**innovative**” and “**effective**,” and have listed my UAV cyberattack detection as one of the commonly used **baseline** categories, indicating that my work becomes a practical reference point for subsequent research and benchmarking.

➤ Key contributions

✓ **C1: Vision-based learning for collaboration: swarm search and rescue**

A first contribution advances **vision-based collaboration** for target search and rescue, spanning **target search (TNNLS)**, **multi-target search (Advanced Intelligent Systems)**, and **heterogeneous air-ground search (MIR)**, complemented by a **novel air search platform (TRO)**. The technical theme is scalable coordination from local perception: agents must explore efficiently, avoid redundant coverage, and share task-relevant information under partial observability and communication limits. By extending from single-target to multi-target objectives and then to heterogeneous UAV-UGV teams, this line establishes a coherent progression from “learning to search” to “learning to coordinate search” to “learning to coordinate across platforms.” The resulting methods provide practical algorithmic ingredients for swarm autonomy in cluttered, dynamic environments.

✓ **C2: Vision-based learning for adversaries: detection, evasion, and resilient control**

A second contribution develops **vision-based resilience under adversaries and failures**, including **multi-pursuit detection and tracking (TIV, CASE 2024)**, **multi-pursuit evasion with DRL (TAI)**, **cyberattack detection and isolation (TVT)**, and **resilient formation control (IoTJ)**. The unifying objective is to maintain safe swarm behavior when threats exist at multiple layers: (i) perception degradation and deception, (ii) strategic non-stationarity from multiple opponents, and (iii) integrity threats in networked sensing/control loops. This work emphasizes robustness and safety as design targets rather than post-hoc evaluation criteria, and it establishes experimentally grounded pathways to “graceful degradation” rather than brittle failure in adversarial settings.

✓ **C3: VLA for planning: structured priors for controllable autonomy**

A third contribution explores **VLA-style planning** to bridge high-level semantic reasoning and low-level control: **FM-Planner (TITS)** and **VL-Fly (IoTJ)** investigate how vision-language representations can provide structured guidance, subgoals, constraints, and scene-grounded cues, to improve planning efficiency and controllability. This direction targets a persistent gap in embodied learning: end-to-end RL can be sample-hungry and hard to constrain, while purely symbolic planning struggles with noisy perception. VLA-inspired planning offers a practical middle layer that improves generalization and enables more interpretable decision pipelines.

➤ **Future research direction and plan: general vision-based learning for swarm intelligence**

Building on C1-C3, the next stage will develop **general vision-based learning for swarm intelligence** with three thrusts that jointly address perception consistency, cross-domain generalization, and safety/resilience.

1) Collaborative perception consistency. Learn **adaptive hierarchical topologies** (dynamic leaders/relays), develop **alignment-ready representations** for **multi-view spatiotemporal fusion**, and design **bandwidth-aware consensus** (selective/event-triggered, compressed messaging) to achieve consistent swarm perception across domains.

2) Collaborative foundation models. Build **multimodal universal encodings** that unify heterogeneous sensors/actions, enable capability-aware on-demand cooperation for dynamic task allocation under budgets, and distill large-model-driven cooperative priors into lightweight deployable swarm policies for fast adaptation.

3) Safety and resilience. Improve deployability via **mechanism-level interpretability**, adversarial stress testing of perception/communication/policies, and resilient policy generation (fallback modes, uncertainty-triggered switching, continual adaptation) for safe operation under shifts and partial failures.

My research aims to provide enabling methods for real-world swarm intelligence in high-stakes applications, where reliability and safety are non-negotiable. The outcomes also offer clear, modular entry points for students in **robotics and embodied intelligence**: (i) representation learning and spatiotemporal alignment in world models, (ii) communication-efficient fusion and topology learning, (iii) safe and explainable VLA, and (iv) safety/resilience layers for multi-agent decision-making. Each module supports publishable outcomes while contributing to a coherent long-term goal: highly autonomous and safe swarm intelligence that can see, coordinate, and withstand adversarial and uncertain conditions in the real world.